CALVIN CHI

CONTACT

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EDUCATION

University of California, Berkeley 2015 - 2020
Ph.D., Computational Biology

INDUSTRY

Sr. Applied Scientist at Amazon

Full-Funnel Agents and Models Mar 2025 - present

- LLM post-training (SFT, GRPO) of GPT OSS 120B and Qwen 3.5 122B models for Ads Agent skills, achieving performance gains of up to 2x on internal evaluations.
- Worked on synthetic data generation strategies of sizes 10k - 50k for training and evaluation.
- Launched the Full Funnel Campaign Agent in closed beta, later becoming Amazon Ads Agent.

Demand Side Platform Aug 2020 - Mar 2025

- Developed and launched science team's first deep learning conversion prediction model, pre-ranking model, and internal bidding simulator.
- Designed and launched bidding exploration algorithm, reducing cost-per-conversion by 20%.

SKILLS

Programming: Python, PySpark, R, SQL
ML/AI frameworks: PyTorch, Strands Agents SDK
Graduate Coursework: RL, NLP, Computer Vision, Convex Optimization

RELEVANT PUBLICATIONS

Quantifying Pacing Regret with Bid Replay Simulation
Calvin Chi, Fabian Wauthier, Yu Gan
Amazon Machine Learning Conference (AMLC, internal), 2025

Proximal Epsilon-Greedy Bidding for Online Advertising
Calvin Chi, Victor Pu, Mingyan Cu, Yu Gan
Amazon Machine Learning Conference (AMLC, internal), 2025

Bipartite graph-based approach for clustering of cell lines by gene expression-drug response associations
Calvin Chi, Yuting Ye, Bin Chen, Haiyan Huang
Bioinformatics 37.17, 2021

Identification of Sjögren's syndrome patient subgroups by clustering of labial salivary gland DNA methylation profiles
Calvin Chi, *et al.*
PLoS One 18.3, 2023

PROJECTS

HLA Allele Imputation with Multitask Deep CNN June 2021

- Developed multitask CNN for imputing HLA alleles from genotype data.
- Achieved 97.6% overall imputation accuracy, beating or achieving similar performance as existing SOTA HLA imputation methods.

Embedding-Augmented Deep CNN for PubMed Journal Recommendation Dec 2018

- Journal detection from PubMed abstract with 415,381 programmatically-collected abstracts.
- Compared multitask and embedding-augmented CNNs with output space of 1,548 journals.
- Best performance when CNN input augmented with topic and impact factor embeddings, with accuracy 23.7% and 90% of true journals in top 60 recommendations.

AWARDS

NSF Graduate Research Fellowship Mar 2017